WEEK 1 · DAY 3

# QPMs

*Semi-Structural Quarterly Projection Models*

**01** **The Workhorse QPM**
Economic structure built for central bank policy analysis

**02** **Unobserved Variables & Judgment**
Potential output, r*, NAIRU, expectations

**03** **Sticky-Price Inflation Framework**
The core of credibility measurement

Chulalongkorn University · Bangkok, Thailand · June 10, 2026

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# What You'll Learn Today

*Where economics meets the data — and where judgment meets the model*

### QPM architecture

The four core equations: output gap, Phillips curve, policy rule, and exchange rate.

### Unobserved variables

Why potential output, r*, and NAIRU must be estimated — and why judgment is essential.

### Sticky & flexible prices

Why sticky-price inflation is the right lens for policy credibility.

### Calibration & scenario design

How model parameters are tailored to a country and set up for Case A/B scenarios.

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# Why Semi-Structural?

*The pragmatic middle ground — economic content without unmanageable complexity*

PURE TIME SERIES

## VAR / Black Box

- Forecasts without structural interpretation
- No clear role for policy or credibility
- Hard to tell scenario stories

SEMI-STRUCTURAL QPM

## The Workhorse

- Small set of behaviorally-meaningful equations
- Clear economic interpretation
- Built for policy trade-off analysis
- Tractable for monthly/quarterly cycles

FULL DSGE

## Structural but Heavy

- Strong microfoundations
- Useful for welfare & deep analysis
- Slow to run; risk of misspecification
- Harder to operate in a policy cycle

*QPMs give forecasting teams economic content they can defend — and simplicity they can actually run every cycle.*

# Mixed Complementarity Problem (MCP) Ingredients

*The policy problem assembled from three pieces — solved jointly with occasionally-binding constraints*

INGREDIENT 01

### Loss Function Minimization

The central bank chooses policy to minimize a weighted sum of deviations — inflation from target, output from potential, and changes in the policy rate.

INGREDIENT 02

### Linear + Nonlinear Equations

The economy is a system of linear and nonlinear equations — including a convex Phillips curve and endogenous policy credibility that responds to performance.

INGREDIENT 03

### Occasionally-Binding Constraints

Nasty constraints enter and exit — most prominently the effective lower bound on interest rates — requiring complementarity-style solution methods.

***Solved jointly as a Mixed Complementarity Problem — the right tool for kinked policy rules and active ELB constraints.***

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# Four Core QPM Equations

*A small, coherent system that captures the transmission mechanism*

### 01 Output Gap (IS)

$$y\,gap = \rho \cdot y\,gap(-1) - \beta \cdot (r - r^*) + \gamma \cdot (z - z^*)$$

Aggregate demand responds to the real interest rate gap and real exchange rate gap.

### 02 Phillips Curve

$$\pi = \alpha \cdot E\pi + (1-\alpha) \cdot \pi(-1) + \lambda \cdot y\,gap + \varepsilon\,\pi$$

Inflation driven by expectations, inertia, and the output gap. Sticky/flexible split applies here.

### 03 Policy Rule (Taylor-type)

$$i = \varphi \cdot i(-1) + (1-\varphi) \cdot [\, i^* + \theta\pi \cdot (\pi - \bar{\pi}) + \theta y \cdot y\,gap \,]$$

The central bank reacts systematically to inflation and output gaps, with smoothing.

### 04 Risk-Adjusted UIP

$$i = \Delta s + i^* + \rho$$

Exchange rate dynamics link domestic policy, foreign rates, and country risk premium.

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# Unobserved Variables

*The starred quantities — y*, r*, u*, π* — are estimated, not measured. Judgment is essential.*

KEY UNOBSERVABLES

$y^*$ **Potential output**
Economy's non-inflationary capacity

$r^*$ **Neutral interest rate**
Rate consistent with stable inflation

$u^*$ **NAIRU**
Unemployment consistent with stable inflation

$\rho$ **Country risk premium**
Wedge between domestic and foreign yields

$E\pi$ **Inflation expectations**
Forward-looking behavior of price-setters

WHY JUDGMENT MATTERS

## Estimates change with every vintage

- No filter or model can identify these from data alone — priors and structural assumptions matter
- Potential output revisions can flip a positive output gap to negative — policy implications shift overnight
- r* is inherently contested; different methods disagree by hundreds of basis points
- Mark II makes these priors explicit and tests robustness across scenarios

# Monetary Policy Loss Function

*The central bank minimizes a discounted sum of three quadratic terms*

$$Loss = \Sigma\ [\ 1.0\cdot(\pi 4_{t+i} - \pi^*)^2 + 1.0\cdot y^2_{t+i} + 0.5\cdot(rs_{t+i} - rs_{t+i-1})^2\ ]$$

*Summed over i = 0 to ∞*

### Under IT, target deviations are costly

Inflation deviations from $\pi^*$ are penalised — because under inflation targeting, the target is the nominal anchor.

### Avoid output deviations

Penalising y²_t minimises the amplitude of the business cycle and prevents prolonged gaps in either direction.

### Gradualism in policy rates

Penalising changes in the real rate (rs − rs_(−1)) reflects aversion to sharp movements and strengthens expectations channel.

### Asymmetric response (despite symmetric loss)

Even though the loss function is symmetric, endogenous credibility produces stronger responses to overshoots than undershoots.

# Expectations-Augmented Phillips Curve

*Inflation driven by expectations, inertia, and a nonlinear output-gap effect*

$$\pi_t = \lambda_1 \cdot \pi 4^e_t + (1 - \lambda_1) \cdot \pi 4_{t-1} + \lambda_2 \cdot [\, y_{t-1} \cdot y_{max} / (y_{max} - y_{t-1}) \,] + \varepsilon^\pi_t$$

*Non-linear output gap effect*

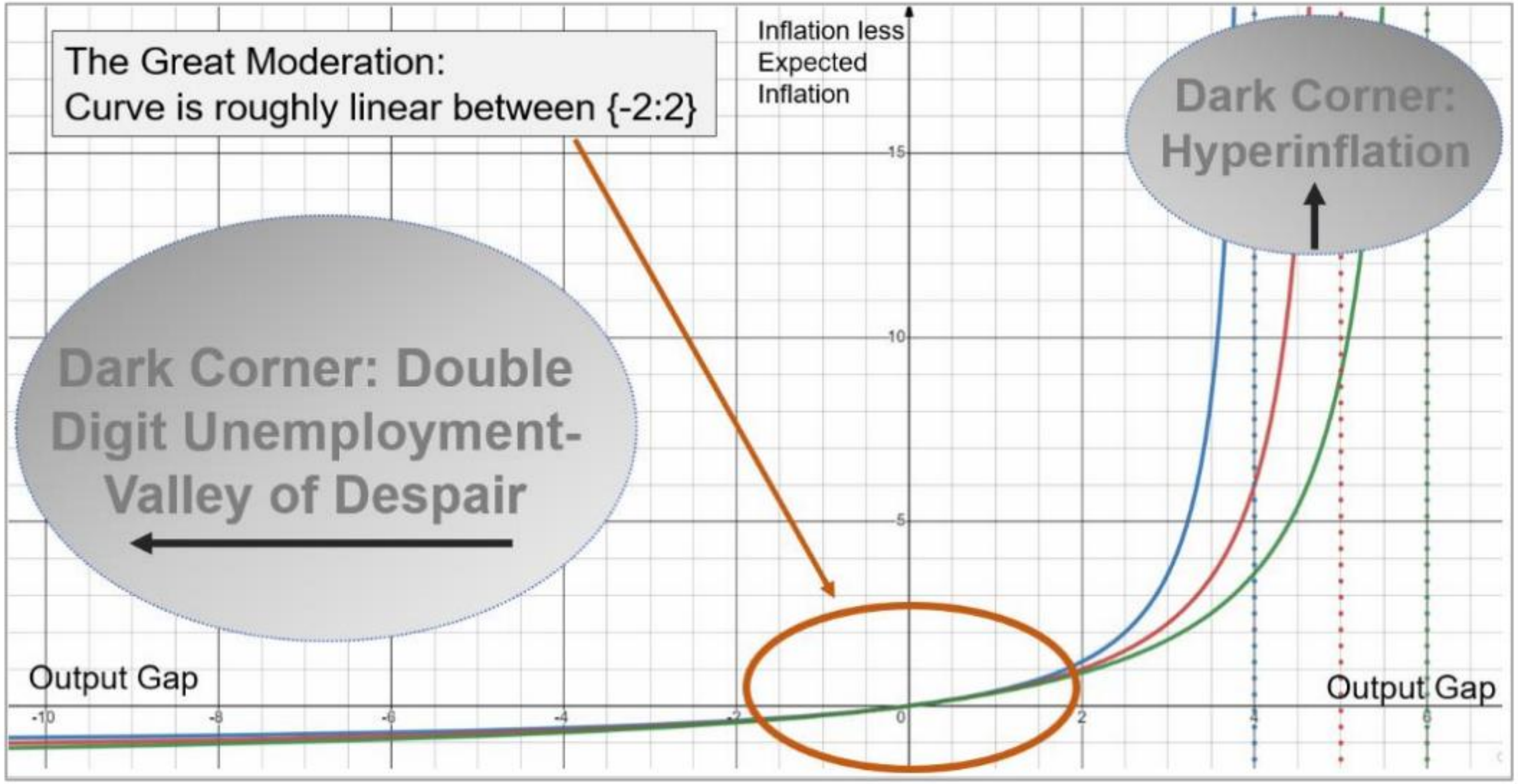
*Calibration:* $\lambda_1 = 0.70$, $\lambda_2 = 0.10$, $y\_max = 6.0$

**WHERE**

- $\pi 4^e_t$ — Inflation expectations
- $\pi 4_{(t-1)} = (1/4)\, \Sigma\, \pi_{(t-i)}$ — Backward-looking component of inflation (4-quarter average)
- $y_{(t-1)}$ — Output gap in period t−1
- $y_{max}$ — Maximum output gap possible before inflation explodes

# Convex Phillips Curve — Different y_max Values

*Non-linear output-gap effect: as y → y_max, the inflation response explodes into the dark corner*




**DARK CORNER · LEFT**

### Double-Digit Unemployment

Valley of despair — large negative gaps, deflationary pressure, monetary policy loses traction at the ELB.

**DARK CORNER · RIGHT**

### Hyperinflation

Capacity constraints bind hard; small further overheating triggers sharp, explosive inflation.

**GREAT MODERATION**

Curve is roughly linear between {−2 : 2} — this is where policy usually operates.

# Inflation Regimes and CB Performance

*Two hypothetical regimes — a Low-inflation anchor and a High-inflation drift*

**LOW INFLATION REGIME (L)**

**Convergence to the π* target**

$$\pi4^L_t = \Gamma^L \cdot \pi4_{t-1} + (1 - \Gamma^L) \cdot \pi^* + \varepsilon^{\pi L}_t$$

Γ^L = 0.5,   π* = 2

**HIGH INFLATION REGIME (H)**

**Convergence to a higher level (10%)**

$$\pi4^H_t = \Gamma^H \cdot \pi4_{t-1} + (1 - \Gamma^H) \cdot 10 + \varepsilon^{\pi H}_t$$

Γ^H = 0.9

CENTRAL BANK PERFORMANCE INDICATOR

$$\eta_t = (\pi4^H_t - \pi4_t)^2 \,/\, [\,(\pi4^H_t - \pi4_t)^2 + (\pi4^L_t - \pi4_t)^2\,]$$

$\eta_t$ = **1** in the **"L"** case · $\eta_t$ = **0** in the **"H"** case

# Why Sticky Prices?

*Credibility is a reflection of performance — and sticky prices are the cleanest signal*

REASON 01

### Anchoring long-term expectations

Sticky prices are "best" for thinking about how well central banks have been anchoring long-term inflation expectations and managing the short-run output-inflation trade-off.

REASON 02

### The price-setting signal

They better reflect the type of prices that are most relevant to central bankers — because they contain information on the underlying price-setting behaviour of the entire economy.

REASON 03

### Non-tradeable services (haircuts)

In a broad sense, sticky prices are the components of the CPI basket that are non-tradeable services. The commonly used example is haircut prices — a non-traded service that requires domestic labour to produce.

*We construct the Central Bank Sticky Price Inflation Indicator (CBSPII) — the basis for measuring performance.*

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# Sticky vs Flexible Price Inflation

*Not all prices tell central banks the same thing — split the CPI to read credibility correctly*

## FLEXIBLE PRICES

### Fast-moving, noisy, weakly informative

- Tradeable goods, commodities, energy, food
- Driven by global prices, exchange rate, weather
- Large, frequent, often transitory movements
- Reading headline inflation alone can mislead policy

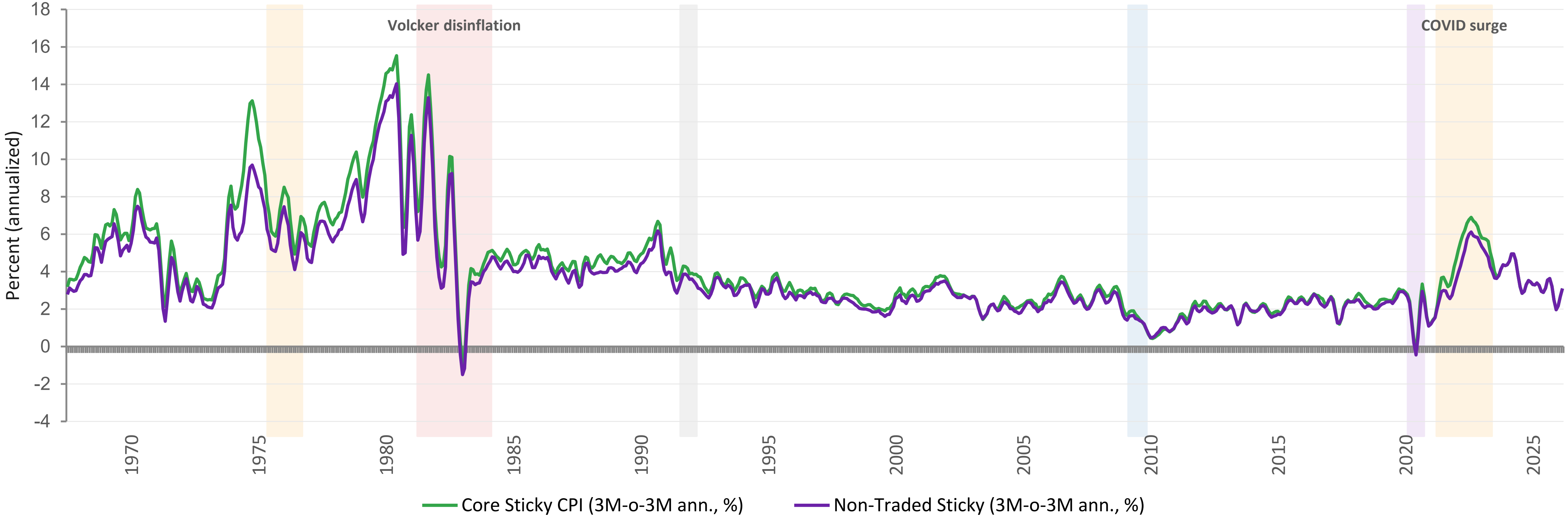
## STICKY PRICES

### Slow-moving, persistent, highly informative

- Non-tradeable services (rent, haircuts, insurance, education)
- Reflect underlying price-setting behaviour of the economy
- Persistence reveals how well expectations are anchored
- The right signal for credibility — used in CBSPII measurement

# US Core Sticky & Non-Traded Sticky Price Inflation

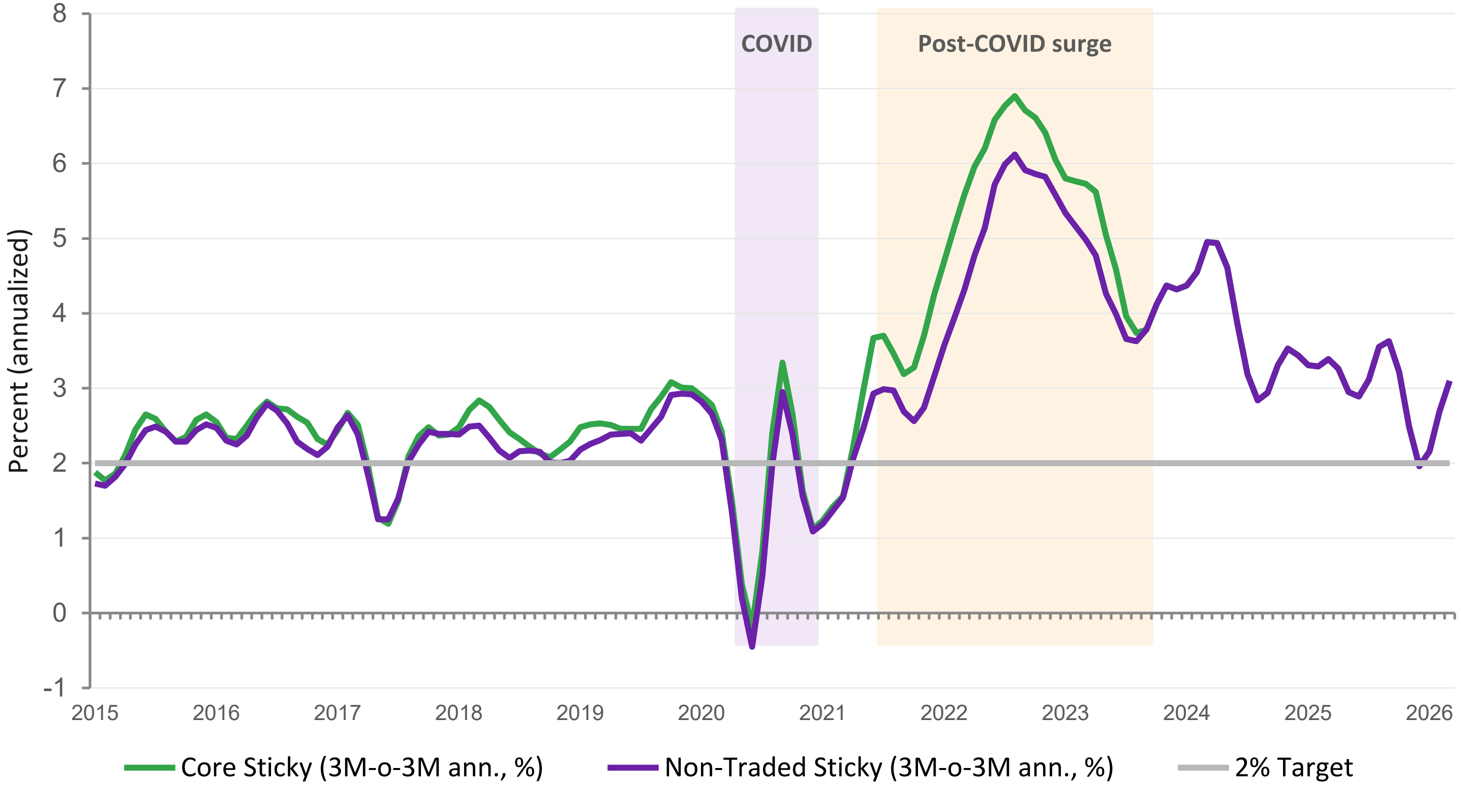
*3-month-over-3-month annualized · 1967–2026 · key episodes shaded*




*Source: Federal Reserve Bank of Atlanta, Core Sticky Price CPI (monthly). Non-Traded Sticky = sticky items with tradeable goods removed (services-focused). 3M-o-3M annualized rate, Feb 1967–Mar 2026.*

# The Post-COVID Episode, 2015–2026

*Core sticky and non-traded sticky — both surged, both still moderating toward 2%*




**CORE STICKY**

## Peaked near 7%

The broadest sticky-price measure hit 6.9% in Aug 2022 — the highest rate since the early 1980s.

**NON-TRADED STICKY**

## Peaked at 6.1%

Stripping out tradeable goods gives a services-heavy view — still tracking above 2% in 2026.

**BOTH**

## Still above target

As of early 2026, both series remain above the 2% target — the classic FPAS-II credibility test.

*Source: Federal Reserve Bank of Atlanta, Core Sticky Price CPI (monthly). Non-Traded Sticky = sticky items with tradeable goods removed. 3M-o-3M annualized.*

# Constructing the Credibility Index — CBCSI

*From performance η_t to the Central Bank Credibility Stock Index γ_t*

## CREDIBILITY STOCK

$$\gamma_t = \rho \cdot \gamma_{t-1} + (1-\rho) \cdot \eta_{t-1} + \varepsilon^{\gamma}_t$$

$\rho = 0.10$ → *credibility accumulates slowly; it's a stock, not a flow.*

## FORMATION OF EXPECTATIONS

$$\pi 4^{e}_t = C_t \cdot \pi 4_{t+4} + (1 - C_t) \cdot \pi 4_{t-1} + b_t + \varepsilon^{\pi e}_t$$

*with inflation-expectation bias* $b_t = 0.1 \cdot (1 - C_t)$

**C_t ∈ [0, 1]**
Credibility is bounded. 1 = fully anchored, 0 = no trust.

**High credibility**
Agents put more weight on the target — expectations stay anchored.

**Low credibility**
Agents weight past inflation more; bias b_t grows.

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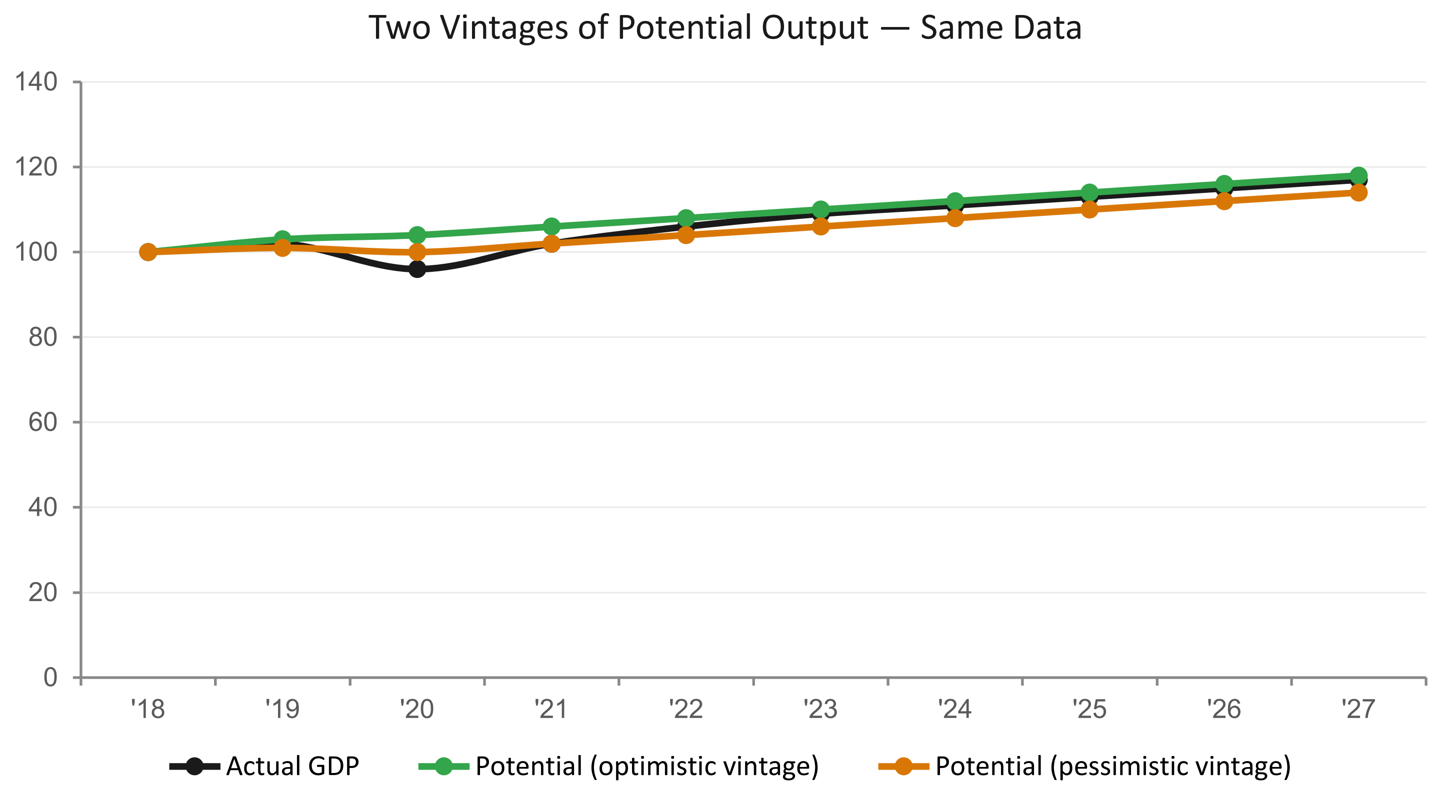
# Output Gap: The Hardest Unobservable

*Potential output estimates drive policy recommendations — and they move with every data vintage*




## IMPLICATIONS FOR POLICY

**01** **Sign of the output gap can flip with revisions**

The same GDP path is inflationary or disinflationary depending on potential.

**02** **Policy rules amplify this uncertainty**

Small changes in y* translate directly into different rate recommendations.

**03** **Scenario approach is the response**

Run Case A/B with alternative y* assumptions — don't commit to one estimate.

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# Calibration Workflow

*From the literature to a country-specific model — with scenario design baked in*

01

## Anchor

Start from literature priors and the country's institutional features (FX regime, openness, financial depth).

02

## Match moments

Tune parameters so model impulse responses and volatilities match stylized facts.

03

## Stress test

Run historical shocks through the model — does it reproduce known episodes?

04

## Design scenarios

Set up Case A / Case B / Market Ref around shared priors, with explicit uncertainty in y* and r*.

***Tomorrow — Day 4 — we move this calibrated model into DynareJulia and run the first scenario simulations.***

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# Discussion & Wrap-up

*Questions to prepare for tomorrow's DynareJulia implementation*

## KEY TAKEAWAYS FROM TODAY

- Semi-structural QPMs sit between atheoretical VARs and heavy DSGEs — enough economics, enough speed.
- Four core equations — output gap, Phillips curve, policy rule, UIP — carry most of the policy content.
- Unobservables ($y^*$, $r^*$, $u^*$, $E\pi$) cannot be read off the data; judgment and priors are part of the model.
- Sticky-price inflation is the right signal for credibility — flexible prices add noise, not information.

## DISCUSSION QUESTIONS

1. How is potential output estimated in your institution today — and how often does it get revised?
2. Does your current policy rule give more weight to inflation or to the output gap? How would you defend that?
3. Which sticky-price indicator would you build for your economy, and what data constraints would you face?
4. If you had to pick one Case A / Case B pair for next quarter, what would differ across them?